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DEPARTMENT OF AI&DS

CGB1122-DATA STRUCTURES

EXPENSE TRACKING SYSTEM

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PROJECT INTRODUCTION



The Expense Tracking System is a simple application designed to help users manage their income and expenses efficiently. It records daily transactions, categorizes expenses, tracks budgets, and generates reports to improve financial planning and spending management.



OBJECTIVE

To develop an Expense Tracking System that helps users record and manage income and expenses, categorize transactions, monitor budgets, and generate financial reports for effective financial planning and control.



DATA STRUCTURES USED



- **Linked Lists**

Useful for dynamic storage of transactions when the number of entries is not fixed.

Allows easy insertion and deletion of records.

- **Stacks**

Can be used for undo operations (e.g., undo last transaction entry).

- **Queues**

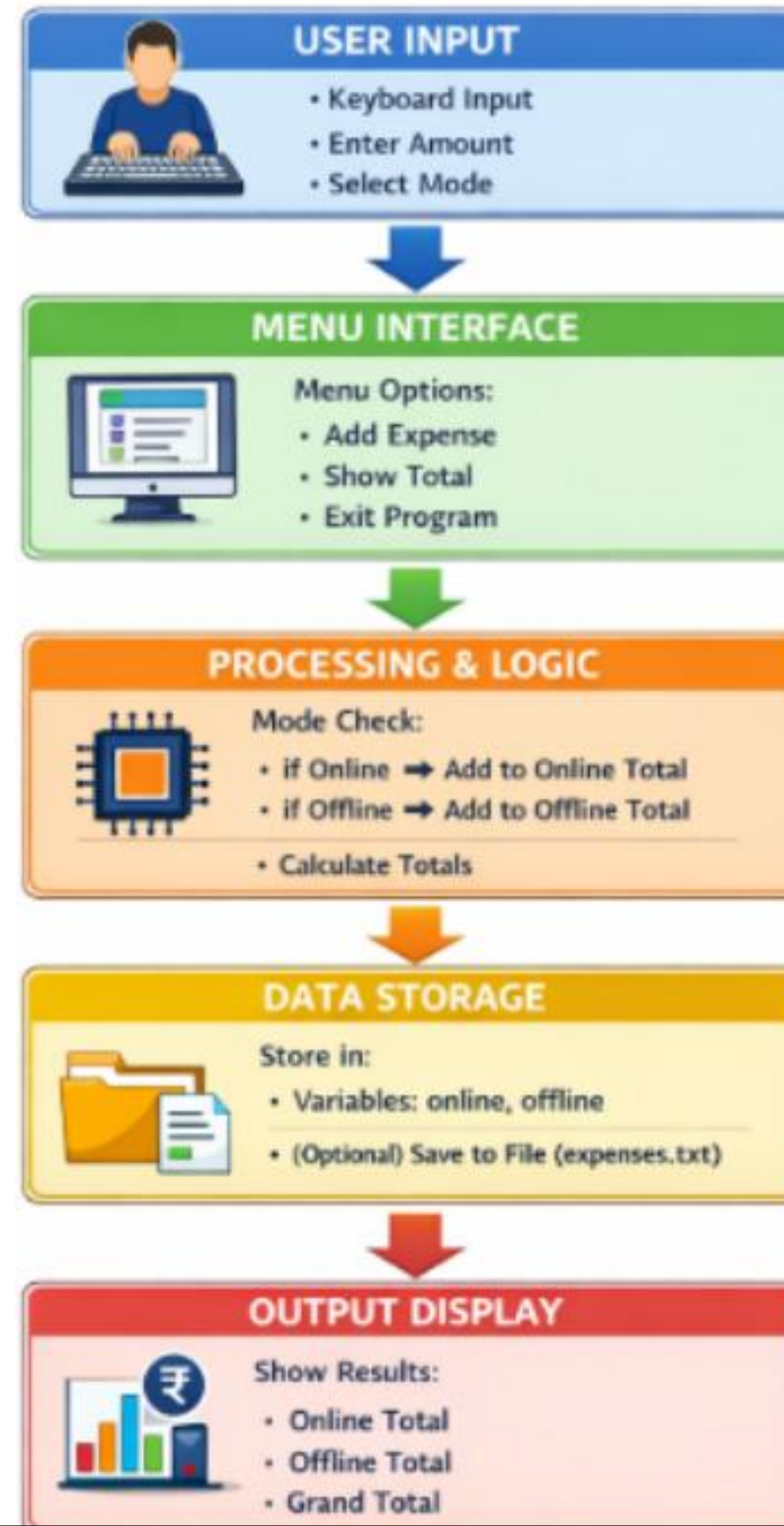
To manage pending transactions or scheduled payments in order.



MODULES USED



ARCHITECTURE DIAGRAM





PROGRAM AND OUTPUT



main.c

```
1 #include <stdio.h>
2 int main() {
3     int n, i, type;
4     float amount, total = 0, cash_total = 0, online_total = 0;
5
6     printf("Enter number of expenses: ");
7     scanf("%d", &n);
8     for(i = 1; i <= n; i++) {
9         printf("\nExpense %d amount: ", i);
10        scanf("%f", &amount);
11        printf("Payment type (1 = Cash, 2 = Online): ");
12        scanf("%d", &type);
13        total += amount;
14        if(type == 1)
15            cash_total += amount;
16        else if(type == 2)
17            online_total += amount;
18        else
19            printf("Invalid type!\n");
20    }
21    printf("\n--- Expense Summary ---\n");
22    printf("Total Expense: %.2f\n", total);
23    printf("Cash Expense: %.2f\n", cash_total);
24    printf("Online Expense: %.2f\n", online_total);
25    return 0;
```

Run

Output

Clear

```
Enter number of expenses: 1

Expense 1 amount: 500
Payment type (1 = Cash, 2 = Online): 2

--- Expense Summary ---
Total Expense: 500.00
Cash Expense: 0.00
Online Expense: 500.00

=== Code Execution Successful ===
```



FUTURE SCOPE



- **Mobile Application Integration** – Develop Android/iOS apps for on-the-go expense tracking.
- **Cloud Synchronization** – Enable data storage and access across multiple devices with real-time updates.
- **AI- Based Insights** – Use machine learning to predict spending patterns and suggest savings strategies.
- **Multi- Currency Support** – Allow users to manage expenses in different currencies for global usability.
- **Bank/Wallet Integration** – Connect with online banking or digital wallets to auto-import transactions.



CONCLUSION



The **Expense Tracking System** provides a reliable and efficient solution for managing personal and organizational finances. By integrating modules such as user authentication, transaction management, expense categorization, budget monitoring, and report generation, the system ensures that users can record, analyze, and control their financial activities with ease.



REFERENCES



- **Books & Academic Sources**
- E. Balagurusamy, *Programming in ANSI C*, McGraw Hill Education.
- Thomas H. Cormen et al., *Introduction to Algorithms*, MIT Press.
- **Web Resources**
- GeeksforGeeks – Tutorials on Data Structures and System Design.
- TutorialsPoint – Concepts of Expense Management Systems.
- ResearchGate – Papers on Financial Management Systems.



THANK YOU!

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Presentation

